

Human review packet: Roth82StableMatching

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: Roth82StableMatching

Paper id: Roth82StableMatching

Source version: Mathematics of Operations Research 7(4), 1982, printed pages 617-628

Source record: audit/paper_statement_map.json

Rows: 10

Generated: 2026-09-06

Source URL: [official source](#)

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper Roth82StableMatching --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

Contents

- [Governing Lean declarations \(29; supporting context\)](#)
- [Paper-specific semantic prerequisites \(12; not paper claims\)](#)
 - [Roth82StableMatching.PaperInterface.completeMarriageOutcome](#)
 - [Roth82StableMatching.PaperInterface.efficientMatchingProcedure](#)
 - [Roth82StableMatching.PaperInterface.fillsIndividualQuotas](#)
 - [Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst](#)
 - [Roth82StableMatching.PaperInterface.preferenceProfile](#)
 - [Roth82StableMatching.PaperInterface.matchingProcedure](#)
 - [Roth82StableMatching.PaperInterface.sourceMarriageModel](#)
 - [Roth82StableMatching.PaperInterface.sourceStableMarriage](#)
 - [Roth82StableMatching.PaperInterface.sourceMenOptimal](#)
 - [Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure](#)
 - [Roth82StableMatching.PaperInterface.sourceWomenOptimal](#)
 - [Roth82StableMatching.PaperInterface.truthfulForAllAgents](#)
- [Source claims \(10\)](#)
 - [theorem1_stable_outcome_existsSpec](#)
 - [theorem2_side_optimal_stable_outcomesSpec](#)
 - [theorem3_no_stable_truthful_procedureSpec](#)
 - [theorem4_efficient_truthful_procedure_existsSpec](#)
 - [theorem5_optimal_side_truthfulSpec](#)
 - [corollary5_1_first_choice_need_not_be_misrepresentedSpec](#)
 - [lemma1_simple_report_same_partnerSpec](#)
 - [lemma2_simple_report_harms_no_manSpec](#)
 - [theorem6_no_outcome_strictly_better_for_all_menSpec](#)
 - [theorem7_later_choice_manipulation_unavoidableSpec](#)

Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.Matching.Assignment

Declaration location: reusable library

Lean declaration body

```
field m_match:
  {M : Type u_1} → {W : Type u_2} → AppliedModelingLib.Matching.Assignment M W → M → Option W

field w_match:
  {M : Type u_1} → {W : Type u_2} → AppliedModelingLib.Matching.Assignment M W → W → Option M

field consistent_m:
  ∀ {M : Type u_1} {W : Type u_2} (self : AppliedModelingLib.Matching.Assignment M W) (m : M) (w : W),
    self.m_match m = some w ↔ self.w_match w = some m
```

AppliedModelingLib.Matching.Assignment.swap

Declaration location: reusable library

Lean declaration body

```
type:
  {M : Type u_1} →
  {W : Type u_2} → AppliedModelingLib.Matching.Assignment M W → AppliedModelingLib.Matching.Assignment W M

value:
  fun {M : Type u_1} {W : Type u_2} (mu : AppliedModelingLib.Matching.Assignment M W) =>
    { m_match := mu.w_match, w_match := mu.m_match,
      consistent_m := AppliedModelingLib.Matching.Assignment.swap_proof_1 mu }
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)

AppliedModelingLib.Matching.DAState

Declaration location: reusable library

Lean declaration body

```
field m_match:
  {M : Type u_3} → {W : Type u_4} → AppliedModelingLib.Matching.DAState M W → M → Option W

field w_match:
  {M : Type u_3} → {W : Type u_4} → AppliedModelingLib.Matching.DAState M W → W → Option M

field m_proposals:
  {M : Type u_3} → {W : Type u_4} → AppliedModelingLib.Matching.DAState M W → M → Finset W

field consistent:
  ∀ {M : Type u_3} {W : Type u_4} (self : AppliedModelingLib.Matching.DAState M W) (m : M) (w : W),
    self.m_match m = some w ↔ self.w_match w = some m
```

AppliedModelingLib.Matching.BestRemainingWoman

Declaration location: reusable library

Lean declaration body

```
type:
  {M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → AppliedModelingLib.Matching.DAState M W → M → W → Prop

value:
  fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (s : AppliedModelingLib.Matching.DAState M W) (m : M) (w : W) =>
    w ∈ s.m_proposals m ∧ 0 ≤ val_m m w ∧ ∀ w' ∈ s.m_proposals m, 0 ≤ val_m m w' → val_m m w' ≤ val_m m w
```

Direct retained dependencies

- [AppliedModelingLib.Matching.DAState](#)

AppliedModelingLib.Matching.IsActiveMan

Declaration location: reusable library

Lean declaration body

```
type:
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → AppliedModelingLib.Matching.DAState M W → M → Prop

value:
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (s : AppliedModelingLib.Matching.DAState M W) (m : M) =>
  s.m_match m = none ∧ ∃ w ∈ s.m_proposals m, 0 ≤ val_m m w
```

Direct retained dependencies

- [AppliedModelingLib.Matching.DAState](#)

AppliedModelingLib.Matching.initialDAState

Declaration location: reusable library

Lean declaration body

```
type:
(M : Type u_3) → (W : Type u_4) → [Fintype W] → AppliedModelingLib.Matching.DAState M W

value:
fun (M : Type u_3) (W : Type u_4) [Fintype W] =>
  { m_match := fun (x : M) => none, w_match := fun (x : W) => none, m_proposals := fun (x : M) => Finset.univ,
    consistent := AppliedModelingLib.Matching.initialDAState._proof_1 M W }
```

Direct retained dependencies

- [AppliedModelingLib.Matching.DAState](#)

AppliedModelingLib.Matching.removeProposal

Declaration location: reusable library

Lean declaration body

```
type:
{M : Type u_1} →
{W : Type u_2} → [DecidableEq M] → [DecidableEq W] → AppliedModelingLib.Matching.DAState M W → M → W → M → Finset W

value:
fun {M : Type u_1} {W : Type u_2} [DecidableEq M] [DecidableEq W] (s : AppliedModelingLib.Matching.DAState M W) (m : M)
  (w : W) (a : M) =>
  if a = m then s.m_proposals m \ {w} else s.m_proposals a
```

Direct retained dependencies

- [AppliedModelingLib.Matching.DAState](#)

AppliedModelingLib.Matching.daStep

Declaration location: reusable library

Lean declaration body

```
type:
{M : Type u_1} →
{W : Type u_2} →
[Fintype M] →
[Fintype W] →
[DecidableEq M] →
[DecidableEq W] →
(M → W → ℝ) →
(W → M → ℝ) → AppliedModelingLib.Matching.DAState M W → AppliedModelingLib.Matching.DAState M W

value:
fun {M : Type u_1} {W : Type u_2} [Fintype M] [Fintype W] [DecidableEq M] [DecidableEq W] (val_m : M → W → ℝ)
  (val_w : W → M → ℝ) (s : AppliedModelingLib.Matching.DAState M W) =>
  have x : Decidable (∃ (m : M), AppliedModelingLib.Matching.IsActiveMan val_m s m) :=
    Classical.propDecidable (∃ (m : M), AppliedModelingLib.Matching.IsActiveMan val_m s m);
  if h : ∃ (m : M), AppliedModelingLib.Matching.IsActiveMan val_m s m then
    let m : M := Classical.choose h;
```

```

have hact : AppliedModelingLib.Matching.IsActiveMan val_m s (Classical.choose h) :=
  AppliedModelingLib.Matching.daStep._proof_1 val_m s h;
have w_exists :  $\exists (w : W)$ , AppliedModelingLib.Matching.BestRemainingWoman val_m s m w :=
  AppliedModelingLib.Matching.exists_best_woman val_m s m hact;
let w : W := Classical.choose w_exists;
have hw_best : AppliedModelingLib.Matching.BestRemainingWoman val_m s m (Classical.choose w_exists) :=
  AppliedModelingLib.Matching.daStep._proof_3 val_m s h;
have new_proposals :  $M \rightarrow \text{Finset } W$  := AppliedModelingLib.Matching.removeProposal s m w;
let m_current : Option M := s.w_match w;
have accepts : Prop :=
  match m_current with
  | none =>  $0 \leq \text{val}_w w m$ 
  | some m' =>  $\text{val}_w w m' < \text{val}_w w m$ ;
have x : Decidable accepts := Classical.propDecidable accepts;
if hacc : accepts then
  let new_w_match :  $W \rightarrow \text{Option } M$  := Function.update s.w_match w (some m);
  let new_m_match :  $M \rightarrow \text{Option } W$  := fun (m'' : M) =>
    if m'' = m then some w else if m_current = some m' then none else s.m_match m'';
  { m_match := new_m_match, w_match := new_w_match, m_proposals := new_proposals,
    consistent := AppliedModelingLib.Matching.daStep._proof_4 val_m s h }
else { m_match := s.m_match, w_match := s.w_match, m_proposals := new_proposals, consistent := s.consistent }
else s

```

Direct retained dependencies

- [AppliedModelingLib.Matching.BestRemainingWoman](#)
- [AppliedModelingLib.Matching.DAState](#)
- [AppliedModelingLib.Matching.IsActiveMan](#)
- [AppliedModelingLib.Matching.removeProposal](#)

AppliedModelingLib.Matching.deferredAcceptanceState

Declaration location: reusable library

Lean declaration body

```

type:
{M : Type u_1} →
{W : Type u_2} →
[Fintype M] →
[Fintype W] →
[DecidableEq M] → [DecidableEq W] → (M → W →  $\mathbb{R}$ ) → (W → M →  $\mathbb{R}$ ) → AppliedModelingLib.Matching.DAState M W

value:
fun {M : Type u_1} {W : Type u_2} [Fintype M] [Fintype W] [DecidableEq M] [DecidableEq W] (val_m : M → W →  $\mathbb{R}$ )
  (val_w : W → M →  $\mathbb{R}$ ) =>
  have max_steps :  $\mathbb{N}$  := Fintype.card M * Fintype.card W;
  List.foldl
    (fun (s : AppliedModelingLib.Matching.DAState M W) (x :  $\mathbb{N}$ ) => AppliedModelingLib.Matching.daStep val_m val_w s)
    (AppliedModelingLib.Matching.initialDAState M W) (List.range max_steps)

```

Direct retained dependencies

- [AppliedModelingLib.Matching.DAState](#)
- [AppliedModelingLib.Matching.daStep](#)
- [AppliedModelingLib.Matching.initialDAState](#)

AppliedModelingLib.Matching.deferredAcceptance

Declaration location: reusable library

Lean declaration body

```

type:
{M : Type u_1} →
{W : Type u_2} →
[Fintype M] →
[Fintype W] →
[DecidableEq M] → [DecidableEq W] → (M → W →  $\mathbb{R}$ ) → (W → M →  $\mathbb{R}$ ) → AppliedModelingLib.Matching.Assignment M W

value:
fun {M : Type u_1} {W : Type u_2} [Fintype M] [Fintype W] [DecidableEq M] [DecidableEq W] (val_m : M → W →  $\mathbb{R}$ )
  (val_w : W → M →  $\mathbb{R}$ ) =>
  { m_match := (AppliedModelingLib.Matching.deferredAcceptanceState val_m val_w).m_match,
    w_match := (AppliedModelingLib.Matching.deferredAcceptanceState val_m val_w).w_match,
    consistent_m := AppliedModelingLib.Matching.deferredAcceptance._proof_1 val_m val_w }

```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [AppliedModelingLib.Matching.DAState](#)
- [AppliedModelingLib.Matching.deferredAcceptanceState](#)

Roth82StableMatching.PaperInterface.ManyToManyQuotaAssignment

Declaration location: paper-local

Lean declaration body

```
field partnersM:
  {M : Type u_1} → {W : Type u_2} → Roth82StableMatching.PaperInterface.ManyToManyQuotaAssignment M W → M → Set W

field partnersW:
  {M : Type u_1} → {W : Type u_2} → Roth82StableMatching.PaperInterface.ManyToManyQuotaAssignment M W → W → Set M

field consistent:
  ∀ {M : Type u_1} {W : Type u_2} (self : Roth82StableMatching.PaperInterface.ManyToManyQuotaAssignment M W) (m : M)
    (w : W), w ∈ self.partnersM m ↔ m ∈ self.partnersW w
```

Roth82StableMatching.PaperInterface.menDeferredAcceptance

Declaration location: paper-local

Lean declaration body

```
type:
  {M : Type u_1} →
  {W : Type u_2} →
  [Fintype M] →
  [Fintype W] →
  [DecidableEq M] → [DecidableEq W] → (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W

value:
  fun {M : Type u_1} {W : Type u_2} [Fintype M] [Fintype W] [DecidableEq M] [DecidableEq W] (val_m : M → W → ℝ)
    (val_w : W → M → ℝ) =>
    AppliedModelingLib.Matching.deferredAcceptance val_m val_w
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [AppliedModelingLib.Matching.deferredAcceptance](#)

Roth82StableMatching.PaperInterface.paretoOptimal

Declaration location: paper-local

Lean declaration body

```
type:
  {M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W → Prop

value:
  fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ)
    (mu : AppliedModelingLib.Matching.Assignment M W) =>
    Roth82StableMatching.PaperInterface.completeMarriageOutcome mu ∧
    → ∃ (nu : AppliedModelingLib.Matching.Assignment M W),
      Roth82StableMatching.PaperInterface.completeMarriageOutcome nu ∧
      (∀ (m : M),
        (match mu.m_match m with
        | none => 0
        | some w => val_m m w) ≤
        (match nu.m_match m with
        | none => 0
        | some w => val_m m w) ∧
      (∀ (w : W),
        (match mu.w_match w with
        | none => 0
        | some m => val_w w m) ≤
        (match nu.w_match w with
        | none => 0
        | some m => val_w w m) ∧
      ((∃ (m : M),
        (match mu.m_match m with
        | none => 0
        | some w => val_m m w) <
        (match nu.m_match m with
```

```

| none => 0
| some w => val_m m w) v
∃ (w : W),
(match mu.w_match w with
| none => 0
| some m => val_w w m) <
match nu.w_match w with
| none => 0
| some m => val_w w m)

```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.completeMarriageOutcome](#)

Roth82StableMatching.PaperInterface.strictMarriageDomain

Declaration location: paper-local

Lean declaration body

```

type:
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → Prop

value:
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ) =>
(∀ (m : M) (w w' : W), val_m m w = val_m m w' → w = w') ∧
(∀ (w : W) (m m' : M), val_w w m = val_w w m' → m = m') ∧
(∀ (m : M) (w : W), 0 < val_m m w) ∧ ∀ (w : W) (m : M), 0 < val_w w m

```

Roth82StableMatching.PaperInterface.strictPreferenceProfile

Declaration location: paper-local

Lean declaration body

```

type:
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → Prop

value:
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ) =>
(∀ (m : M) (w w' : W), val_m m w = val_m m w' → w = w') ∧ ∀ (w : W) (m m' : M), val_w w m = val_w w m' → m = m'

```

Roth82StableMatching.PaperInterface.unstableMarriage

Declaration location: paper-local

Lean declaration body

```

type:
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W → Prop

value:
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ)
(mu : AppliedModelingLib.Matching.Assignment M W) =>
∃ (m : M) (w : W),
(match mu.m_match m with
| none => 0
| some w' => val_m m w') <
val_m m w ∧
(match mu.w_match w with
| none => 0
| some m' => val_w w m') <
val_w w m

```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)

Roth82StableMatching.PaperInterface.womenDeferredAcceptance

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} →
{W : Type u_2} →
[Fintype M] →
[Fintype W] →
[DecidableEq M] → [DecidableEq W] → (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W

value:
fun {M : Type u_1} {W : Type u_2} [Fintype M] [Fintype W] [DecidableEq M] [DecidableEq W] (val_m : M → W → ℝ)
  (val_w : W → M → ℝ) =>
  (AppliedModelingLib.Matching.deferredAcceptance val_w val_m).swap
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [AppliedModelingLib.Matching.Assignment.swap](#)
- [AppliedModelingLib.Matching.deferredAcceptance](#)

Roth82StableMatching.Theorem3Agent

Declaration location: paper-local

Lean declaration body

```
type:
Type

value:
Fin 3
```

Roth82StableMatching.paper_matching_valM

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → M → Option W → ℝ

value:
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (m : M) (w : Option W) =>
  match w with
  | none => 0
  | some w' => val_m m w'
```

Roth82StableMatching.paper_matching_valW

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} → {W : Type u_2} → (W → M → ℝ) → W → Option M → ℝ

value:
fun {M : Type u_1} {W : Type u_2} (val_w : W → M → ℝ) (w : W) (m : Option M) =>
  match m with
  | none => 0
  | some m' => val_w w m'
```

Roth82StableMatching.paper_is_stable

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W → Prop

value:
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ)
  (mu : AppliedModelingLib.Matching.Assignment M W) =>
  (∀ (m : M), 0 ≤ Roth82StableMatching.paper_matching_valM val_m m (mu.m_match m)) ∧
  (∀ (w : W), 0 ≤ Roth82StableMatching.paper_matching_valW val_w w (mu.w_match w)) ∧
  ∀ (m : M) (w : W),
    Roth82StableMatching.paper_matching_valM val_m m (mu.m_match m) < val_m m w →
    Roth82StableMatching.paper_matching_valW val_w w (mu.w_match w) < val_w w m → False
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.paper_matching_valM](#)
- [Roth82StableMatching.paper_matching_valW](#)

Roth82StableMatching.paper_profitable_man_misreport

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} →
{W : Type u_2} →
[DecidableEq M] →
((M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) →
(M → W → ℝ) → (W → M → ℝ) → M → (W → ℝ) → Prop

value:
fun {M : Type u_1} {W : Type u_2} [DecidableEq M]
(mechanism : (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) (val_m : M → W → ℝ)
(val_w : W → M → ℝ) (m : M) (report_m : W → ℝ) =>
Roth82StableMatching.paper_matching_valM val_m m ((mechanism val_m val_w).m_match m) <
Roth82StableMatching.paper_matching_valM val_m m ((mechanism (Function.update val_m m report_m) val_w).m_match m)
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.paper_matching_valM](#)

Roth82StableMatching.paper_profitable_woman_misreport

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} →
{W : Type u_2} →
[DecidableEq W] →
((M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) →
(M → W → ℝ) → (W → M → ℝ) → W → (M → ℝ) → Prop

value:
fun {M : Type u_1} {W : Type u_2} [DecidableEq W]
(mechanism : (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) (val_m : M → W → ℝ)
(val_w : W → M → ℝ) (w : W) (report_w : M → ℝ) =>
Roth82StableMatching.paper_matching_valW val_w w ((mechanism val_m val_w).w_match w) <
Roth82StableMatching.paper_matching_valW val_w w ((mechanism val_m (Function.update val_w w report_w)).w_match w)
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.paper_matching_valW](#)

Roth82StableMatching.paper_rank_of_choice

Declaration location: paper-local

Lean declaration body

```
type:
{A : Type u_1} → [Fintype A] → [DecidableEq A] → (A → ℝ) → A → ℕ

value:
fun {A : Type u_1} [Fintype A] [DecidableEq A] (score : A → ℝ) (a : A) => {b : A | score a < score b}.card + 1
```

Roth82StableMatching.paper_report_misrepresents_kth_choice

Declaration location: paper-local

Lean declaration body

```
type:
{A : Type u_1} → [Fintype A] → [DecidableEq A] → (A → ℝ) → (A → ℝ) → ℕ → Prop

value:
fun {A : Type u_1} [Fintype A] [DecidableEq A] (true_score report_score : A → ℝ) (k : ℕ) =>
  ∃ (a : A),
    Roth82StableMatching.paper_rank_of_choice true_score a = k ∧
    Roth82StableMatching.paper_rank_of_choice report_score a ≠ k
```

Direct retained dependencies

- [Roth82StableMatching.paper_rank_of_choice](#)

Roth82StableMatching.paper_strict_marriage_domain

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → Prop

value:
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ) =>
  (∀ (m : M) (w w' : W), val_m m w = val_m m w' → w = w') ∧
  (∀ (w : W) (m m' : M), val_w w m = val_w w m' → m = m') ∧
  (∀ (m : M) (w : W), 0 < val_m m w) ∧ ∀ (w : W) (m : M), 0 < val_w w m
```

Roth82StableMatching.paper_stable_matching_procedure_on_strict_marriage_domain

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} → {W : Type u_2} → ((M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) → Prop

value:
fun {M : Type u_1} {W : Type u_2}
  (mechanism : (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) =>
  ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
    Roth82StableMatching.paper_strict_marriage_domain val_m val_w →
    Roth82StableMatching.paper_is_stable val_m val_w (mechanism val_m val_w)
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.paper_is_stable](#)
- [Roth82StableMatching.paper_strict_marriage_domain](#)

Roth82StableMatching.paper_stable_procedure_has_profitable_strict_kth_choice_misreport

Declaration location: paper-local

Lean declaration body

```
type:
{M : Type u_1} →
{W : Type u_2} →
[Fintype M] →
[Fintype W] →
[DecidableEq M] →
[DecidableEq W] → ℕ → ((M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) → Prop

value:
fun {M : Type u_1} {W : Type u_2} [Fintype M] [Fintype W] [DecidableEq M] [DecidableEq W] (k : ℕ)
  (mechanism : (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) =>
  ∃ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
    Roth82StableMatching.paper_strict_marriage_domain val_m val_w ∧
    ((∃ (m : M) (report_m : W → ℝ),
      Roth82StableMatching.paper_strict_marriage_domain (Function.update val_m m report_m) val_w ∧
      Roth82StableMatching.paper_report_misrepresents_kth_choice (val_m m) report_m k ∧
      Roth82StableMatching.paper_profitable_man_misreport mechanism val_m val_w m report_m) ∨
    ∃ (w : W) (report_w : M → ℝ),
      Roth82StableMatching.paper_strict_marriage_domain val_m (Function.update val_w w report_w) ∧
      Roth82StableMatching.paper_report_misrepresents_kth_choice (val_w w) report_w k ∧
      Roth82StableMatching.paper_profitable_woman_misreport mechanism val_m val_w w report_w)
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.paper_profitable_man_misreport](#)
- [Roth82StableMatching.paper_profitable_woman_misreport](#)
- [Roth82StableMatching.paper_report_misrepresents_kth_choice](#)
- [Roth82StableMatching.paper_strict_marriage_domain](#)

Roth82StableMatching.paper_no_stable_procedure_avoids_strict_kth_choice_manipulation_on

Declaration location: paper-local

Lean declaration body

```
type:
(M : Type u_1) → (W : Type u_2) → [Fintype M] → [Fintype W] → [DecidableEq M] → [DecidableEq W] → ℕ → Prop

value:
fun (M : Type u_1) (W : Type u_2) [Fintype M] [Fintype W] [DecidableEq M] [DecidableEq W] (k : ℕ) =>
  ∀ (mechanism : (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W),
    Roth82StableMatching.paper_stable_matching_procedure_on_strict_marriage_domain mechanism →
      Roth82StableMatching.paper_stable_procedure_has_profitable_strict_kth_choice_misreport k mechanism
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.paper_stable_matching_procedure_on_strict_marriage_domain](#)
- [Roth82StableMatching.paper_stable_procedure_has_profitable_strict_kth_choice_misreport](#)

Paper-specific semantic prerequisites

Roth82StableMatching.PaperInterface.completeMarriageOutcome

Paper-local declaration:

Roth82StableMatching.PaperInterface.completeMarriageOutcome

Source locator: cited publication, lines 140-145

Verbatim paper-source connection

An outcome of the (monogamous) marriage problem is a onemen and women, i.e., an invertible function $x: M \rightarrow W$. An outco

by $x = [(m_1, x(m_1)), (m_2, x(m_2)), \dots, (m_n, x(m_n))]$ where $x(m_i)$

matched with man m_i , and $x^{-1}(w_j) = m_i$ is the man matched wit

A matching x is stable if no man and no woman who are not m

Lean-expanded paper semantic target

type:

$\{M : \text{Type } u_1\} \rightarrow \{W : \text{Type } u_2\} \rightarrow \text{AppliedModelingLib.Matching.Assignment } M \ W \rightarrow \text{Prop}$

value:

$\text{fun } \{M : \text{Type } u_1\} \{W : \text{Type } u_2\} (\mu : \text{AppliedModelingLib.Matching.Assignment } M \ W) \Rightarrow$
 $(\forall (m : M), \exists (w : W), \mu.m_match \ m = \text{some } w) \wedge \forall (w : W), \exists (m : M), \mu.w_match \ w = \text{some } m$

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Section 2 defines a marriage outcome as an invertible one-to-one matching of the equal sides. Assignment supplies mutual consistency, and completeMarriageOutcome requires every man and woman to have a partner, exactly the source outcome condition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.efficientMatchingProcedure

Paper-local declaration:

Roth82StableMatching.PaperInterface.efficientMatchingProcedure

Source locator: cited publication, lines 433-435

Verbatim paper-source connection

To see the role played by stability in Theorem 3, observe that there are efficient matching procedures, i.e., procedures which always yield Pareto optimal (but not necessarily stable) outcomes, which do not give players any incentive to misrepresent

Lean-expanded paper semantic target

type:
{M : Type u_1} → {W : Type u_2} → ((M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) → Prop

value:
fun {M : Type u_1} {W : Type u_2}
 (mechanism : (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) =>
 ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
 Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
 Roth82StableMatching.PaperInterface.paretoOptimal val_m val_w (mechanism val_m val_w)

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.paretoOptimal](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Section 4 calls efficient procedures those always yielding Pareto-optimal, not necessarily stable, outcomes. The target quantifies over legal strict marriage reports and invokes paretoOptimal, whose definition includes a complete outcome and excludes a complete outcome weakly improving all agents and strictly improving one.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.fillsIndividualQuotas

Paper-local declaration:

Roth82StableMatching.PaperInterface.fillsIndividualQuotas

Source locator: cited publication, lines 167-175

Verbatim paper-source connection

The marriage problem as outlined above differs from the general matching problem in three principal respects. First, in the marriage problem, each agent is to be matched

with exactly one partner, but in the general matching problem, different agents may need to be matched with different numbers of partners, so that each agent has a "quota," and an outcome is a function which matches each agent with his quota of partners.⁶ For our purposes, this difference between the marriage problem and the general matching problem turns out to be of no consequence, since all the arguments which will be used can be carried over virtually unchanged to the general case. Second,

Lean-expanded paper semantic target

```
type:
{M : Type u_1} →
{W : Type u_2} → (M → ℕ) → (W → ℕ) → Roth82StableMatching.PaperInterface.ManyToManyQuotaAssignment M W → Prop
```

```
value:
fun {M : Type u_1} {W : Type u_2} (quotaM : M → ℕ) (quotaW : W → ℕ)
(mu : Roth82StableMatching.PaperInterface.ManyToManyQuotaAssignment M W) =>
(∀ (m : M), (mu.partnersM m).Finite ∧ (mu.partnersM m).ncard = quotaM m) ∧
∀ (w : W), (mu.partnersW w).Finite ∧ (mu.partnersW w).ncard = quotaW w
```

Direct retained dependencies

- [Roth82StableMatching.PaperInterface.ManyToManyQuotaAssignment](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Roth's general matching extension assigns every agent its individual quota of partners. The carrier's pair-consistency field records reciprocal matches; fillsIndividualQuotas then requires each finite partner set to have exactly the stated quota on both sides.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst

Paper-local declaration:

Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst

Source locator: cited publication, lines 522-529

Verbatim paper-source connection

We first show that only a certain kind of simple misrepresentation need

considered, since if any successful manipulation is possible, then it can be achieved a simple misrepresentation. Specifically, if $P'(m_i)$ is an arbitrary misrepresentation discussed above, then an equivalent simple misrepresentation $P''(m_i)$ is one in which

$y(m_i)P''(m_i)w_j$ for all $w_j \# y(m_i)$. That is, $P''(m_i)$ is a preference relation which has $y(m_i)$ as the most preferred match of m_i . The justification for calling $P''(m_i)$ an

Lean-expanded paper semantic target

type:
 $\{W : \text{Type } u_1\} \rightarrow (W \rightarrow \mathbb{R}) \rightarrow W \rightarrow \text{Prop}$

value:
 $\text{fun } \{W : \text{Type } u_1\} \text{ (report_m : } W \rightarrow \mathbb{R}) \text{ (wstar : } W) \Rightarrow \forall (w : W), w \neq \text{wstar} \rightarrow \text{report_m } w < \text{report_m } \text{wstar}$

Recorded source-to-declaration screening **Verdict:** matches

Reason: The Section 5 simple misrepresentation P'' ranks the partner $y(m_i)$ obtained under P' above every other possible partner. The target is precisely that strict top-ranking condition for report_m and wstar .

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.preferenceProfile

Paper-local declaration:

Roth82StableMatching.PaperInterface.preferenceProfile

Source locator: cited publication, lines 130-138

Verbatim paper-source connection

defined on W . When no confusion will occur, $P(m_i)$ will sometimes be referred to as the preference profile.

thus the statement " w_k is preferred by m_i to w ." can be written as $w_k \succ_{m_i} w$

Each w_i in W has a similar preference $P(w_i)$ defined on M .

$= (P(m_1), \dots, P(m_n))$ the $2n$ -vector of all the agents' preferences.

Lean-expanded paper semantic target

type:
 $\text{Type } u \rightarrow \text{Type } v \rightarrow \text{Type } (m \succ w)$

value:
 $\text{fun } (M : \text{Type } u) (W : \text{Type } v) =>$
 $\{ \text{profile} : (M \rightarrow W \rightarrow \mathbb{R}) \times (W \rightarrow M \rightarrow \mathbb{R}) //$
 $(\forall (m : M) (w w' : W), \text{profile.1 } m \ w = \text{profile.1 } m \ w' \rightarrow w = w') \wedge$
 $\forall (w : W) (m m' : M), \text{profile.2 } w \ m = \text{profile.2 } w \ m' \rightarrow m = m' \}$

Recorded source-to-declaration screening Verdict: matches

Reason: Section 2 defines the preference profile as all agents' complete, transitive, strict ordinal relations. A real-valued score for each opposite-side agent with no ties induces exactly such a strict ranking on the finite marriage domain, and the target carries one score family for each side.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.matchingProcedure

Paper-local declaration:

Roth82StableMatching.PaperInterface.matchingProcedure

Source locator: cited publication, lines 300-303

Verbatim paper-source connection

4. Incentives and stability. Since each agent alone knows his own preferences, any matching procedure which depends on agents' preferences can be thought of as consisting of two parts: a mechanism for eliciting the preferences of the agents, and a mechanism for aggregating these elicited preferences to determine an outcome. This

Lean-expanded paper semantic target

type:

Type u → Type v → Type (max u v)

value:

fun (M : Type u) (W : Type v) =>

```
Roth82StableMatching.PaperInterface.preferenceProfile M W →  
{ mu : AppliedModelingLib.Matching.Assignment M W //  
  Roth82StableMatching.PaperInterface.completeMarriageOutcome mu }
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.completeMarriageOutcome](#)
- [Roth82StableMatching.PaperInterface.preferenceProfile](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The paper treats a matching procedure as mapping a reported preference profile to an outcome. The target takes Roth's strict two-sided profile carrier to an Assignment bundled with completeMarriageOutcome, preserving the marriage-model outcome requirement.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.sourceMarriageModel

Paper-local declaration:

Roth82StableMatching.PaperInterface.sourceMarriageModel

Source locator: cited publication, lines 176-197

Verbatim paper-source connection

the marriage problem was defined as having equal numbers of men and women, and without the possibility that a feasible outcome could leave any agent unmatched, while in the general matching problem, there may be an excess of one kind of agent, and an

outcome need not make a match for every agent. This more general case can be handled by adding suitable "dummy" agents corresponding to the option of being unmatched in the final outcome. The third respect in which the marriage problem as described above is more restrictive than the general matching problem is that indiffer-

ence between potential partners has been ruled out by the assumption that all preferences are strict. Relaxing this assumption would actually complicate some of the results. To avoid these complications, only strict preferences will be considered.

Thus the marriage problem will be used to represent the general matching problem with strict preferences. This should not obscure the fact that much of the interest of the 5That is, the preference relations $P(m_i)$ (or $P(w_i)$) have the following properties:

(1) transitivity: if $w_j P_i w_k$ and $w_k P_i w_l$, then $w_j P_i w_l$;

(2) completeness: for all distinct w_j, w_k in W , exactly one of the relations $w_j P_i w_k$ or $w_k P_i w_j$ holds

The interpretation of the preference relations P_i as strict preference relations means that we are ruling out the possibility that an agent will be indifferent between two potential partners. (Also, it is never the case that $w_j P_i w_l$.) See the discussion at the end of this section.

Lean-expanded paper semantic target

type:

```
{M : Type u_1} → {W : Type u_2} → [Fintype M] → [Fintype W] → (M → W → ℝ) → (W → M → ℝ) → Prop
```

value:

```
fun {M : Type u_1} {W : Type u_2} [Fintype M] [Fintype W] (val_m : M → W → ℝ) (val_w : W → M → ℝ) =>
  Fintype.card M = Fintype.card W ∧ Roth82StableMatching.PaperInterface.strictPreferenceProfile val_m val_w
```

Direct retained dependencies

- [Roth82StableMatching.PaperInterface.strictPreferenceProfile](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Roth's marriage model has equal finite sets M and W and strict preferences over the opposite side. sourceMarriageModel states equal cardinality together with the strict two-sided score representation of those rankings.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.sourceStableMarriage

Paper-local declaration:

Roth82StableMatching.PaperInterface.sourceStableMarriage

Source locator: cited publication, lines 145-165

Verbatim paper-source connection

A matching x is stable if no man and no woman who are not matched at x prefer each other to their partners at x . That is, x is stable if there is no man m and woman w such that both

(i) m prefers w to his partner at x , and

(ii) w prefers m to her partner at x .

If a pair m and w do exist satisfying (i) and (ii), then x is not stable.

The motivation for this terminology is that the set of stable outcomes is equal to the set of outcomes that are not blocked by any man and woman whose preference for an outcome is determined by their preferences for their partners.

Lean-expanded paper semantic target

```
type:
  {M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W → Prop
value:
  fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ)
    (mu : AppliedModelingLib.Matching.Assignment M W) =>
    Roth82StableMatching.PaperInterface.completeMarriageOutcome mu ∧
    ¬Roth82StableMatching.PaperInterface.unstableMarriage val_m val_w mu
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.completeMarriageOutcome](#)
- [Roth82StableMatching.PaperInterface.unstableMarriage](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Section 2 defines stability as a complete marriage outcome with no unmatched man-woman pair who both prefer each other to their assigned partners. sourceStableMarriage is completeMarriageOutcome together with the negation of precisely that blocking-pair predicate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.sourceMenOptimal

Paper-local declaration:

Roth82StableMatching.PaperInterface.sourceMenOptimal

Source locator: cited publication, lines 463-466

Verbatim paper-source connection

results, which make use of the fact that Theorem 1 permits us to identify, for each set

of agents, a unique optimal stable outcome, which they each like at least as well as any other stable outcome.

Lean-expanded paper semantic target

type:

```
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W → Prop
```

value:

```
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ)
  (mu : AppliedModelingLib.Matching.Assignment M W) =>
  Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu ∧
  ∀ (mu' : AppliedModelingLib.Matching.Assignment M W),
  Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu' →
  ∀ (m : M),
  (match mu'.m_match m with
  | none => 0
  | some w => val_m m w) ≤
  match mu.m_match m with
  | none => 0
  | some w => val_m m w
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.sourceStableMarriage](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Theorem 2's source definition of a men-optimal stable outcome requires stability and weak preference by every man over every other stable outcome. sourceMenOptimal requires source stability and compares every complete source-stable marriage to mu using each man's true score in exactly that direction.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure

Paper-local declaration:

Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure

Source locator: cited publication, lines 313-323

Verbatim paper-source connection

A different kind of problem concerning incentives arises for matching procedures which do not result in stable outcomes, since such procedures give at least one pair of agents the incentive to disregard the matching procedure, and seek an alternative outcome. Of course, it may be possible to compel the agents to accept the outcome, in spite of these incentives. For instance, the procedure by which some high school athletes are matched with colleges involves signing "letters of intent," after which athletes are effectively prohibited from further negotiating with other schools. (Professional athletes in several sports are matched with teams under an even more restrictive draft, which prevents a player from negotiating with any team but the one which drafts him.) However in situations in which compulsion plays no part, it is desirable for a matching procedure to yield stable outcomes. A procedure which does so for arbitrary preference profiles will be called a stable matching procedure.

Lean-expanded paper semantic target

type:
{M : Type u_1} → {W : Type u_2} → ((M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) → Prop

value:
fun {M : Type u_1} {W : Type u_2}
 (mechanism : (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) =>
 ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
 Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
 Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w (mechanism val_m val_w)

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.sourceStableMarriage](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Section 4 defines a stable matching procedure as one that yields a stable outcome for arbitrary preference profiles. The target requires every legal strict profile to map to sourceStableMarriage, including the source's complete outcome and no-blocking-pair condition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.sourceWomenOptimal

Paper-local declaration:

Roth82StableMatching.PaperInterface.sourceWomenOptimal

Source locator: cited publication, lines 463-466

Verbatim paper-source connection

results, which make use of the fact that Theorem 1 permits us to identify, for each set

of agents, a unique optimal stable outcome, which they each like at least as well as any other stable outcome.

Lean-expanded paper semantic target

type:

```
{M : Type u_1} → {W : Type u_2} → (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W → Prop
```

value:

```
fun {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ)
  (mu : AppliedModelingLib.Matching.Assignment M W) =>
  Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu ∧
  ∀ (mu' : AppliedModelingLib.Matching.Assignment M W),
  Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu' →
  ∀ (w : W),
  (match mu'.w_match w with
   | none => 0
   | some m => val_w w m) ≤
  (match mu.w_match w with
   | none => 0
   | some m => val_w w m)
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.sourceStableMarriage](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The women-optimal counterpart of Theorem 2 requires stability and weak preference by every woman over every other stable outcome. sourceWomenOptimal is the symmetric complete source-stable comparison using each woman's true score.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Roth82StableMatching.PaperInterface.truthfulForAllAgents

Paper-local declaration:

Roth82StableMatching.PaperInterface.truthfulForAllAgents

Source locator: cited publication, lines 340-351

Verbatim paper-source connection

Before proceeding further with this discussion, we need to make precise what is meant by a procedure which gives agents an incentive to reveal their true preferences. Once a given mechanism for aggregating stated preferences is adopted, the matching problem becomes a noncooperative game among the agents, whose payoff consists of the outcome which results, and whose strategy choices consist of what preferences to state (or act according to, cf. footnote 9). Here we define a procedure which gives the agents the incentives to reveal their true preferences as one which aggregates stated

preferences in a manner such that, in the resulting noncooperative game, it is a dominant strategy for each player to state his preferences honestly.¹⁰ In such a procedure, no matter what preferences other players may state, a player who misrepresents his preferences can achieve no better outcome than if he had
↪ stated them correctly, and he may, of course, do worse."

Lean-expanded paper semantic target

```
type:
{M : Type u_1} →
{W : Type u_2} →
[DecidableEq M] → [DecidableEq W] → ((M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) → Prop

value:
fun {M : Type u_1} {W : Type u_2} [DecidableEq M] [DecidableEq W]
(mechanism : (M → W → ℝ) → (W → M → ℝ) → AppliedModelingLib.Matching.Assignment M W) =>
(∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
∀ (m : M) (report_m : W → ℝ),
Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
(match (mechanism (Function.update val_m m report_m) val_w).m_match m with
| none => 0
| some w => val_m m w) ≤
match (mechanism val_m val_w).m_match m with
| none => 0
| some w => val_m m w) ∧
∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
∀ (w : W) (report_w : M → ℝ),
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w) →
(match (mechanism val_m (Function.update val_w w report_w)).w_match w with
| none => 0
| some m => val_w w m) ≤
match (mechanism val_m val_w).w_match w with
| none => 0
| some m => val_w w m)
```

Direct retained dependencies

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Section 4 defines truthful revelation as a dominant strategy: regardless of the other reports, a unilateral misreport cannot produce a better true outcome. truthfulForAllAgents gives exactly those two one-side-at-a-time weak inequalities over legal reported profiles for men and women.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 227

Verbatim source input

THEOREM 1. The set of stable outcomes is always nonempty.

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [DecidableEq M] [DecidableEq W]
(val_m : M → W → ℝ) (val_w : W → M → ℝ),
Fintype.card M = Fintype.card W →
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
∃ (mu : AppliedModelingLib.Matching.Assignment M W),
Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu
```

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.sourceStableMarriage](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)

Lean-elaborated claim roles

1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: Fintype M
4. parameter: Fintype W
5. parameter: DecidableEq M
6. parameter: DecidableEq W
7. parameter: M → W → ℝ
8. parameter: W → M → ℝ
9. assumption: Fintype.card M = Fintype.card W
10. assumption: Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w
11. conclusion: ∃ (mu : AppliedModelingLib.Matching.Assignment M W), Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.theorem1_stable_outcome_exists

Recorded source-to-Spec screening Verdict: matches

Reason: Theorem 1 states that the set of stable outcomes is always nonempty and its proof constructs one for arbitrary marriage preferences. The target quantifies over finite equal-sided strict marriage profiles and returns an assignment satisfying the source-stable predicate, whose definition includes complete matching and no blocking pair.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 258-259

Verbatim source input

THEOREM 2. There is a stable outcome weakly preferred by every man to any other stable outcome, and one weakly preferred by every woman.

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [DecidableEq M] [DecidableEq W]
(val_m : M → W → ℝ) (val_w : W → M → ℝ),
Fintype.card M = Fintype.card W →
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
  (∃ (mu : AppliedModelingLib.Matching.Assignment M W),
    Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w mu) ∧
  (∃ (nu : AppliedModelingLib.Matching.Assignment M W),
    Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w nu)
```

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.sourceMenOptimal](#)
- [Roth82StableMatching.PaperInterface.sourceWomenOptimal](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: Fintype M
4. parameter: Fintype W
5. parameter: DecidableEq M
6. parameter: DecidableEq W
7. parameter: M → W → ℝ
8. parameter: W → M → ℝ
9. assumption: Fintype.card M = Fintype.card W
10. assumption: Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w
11. conclusion: (∃ (mu : AppliedModelingLib.Matching.Assignment M W),
  Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w mu) ∧
  (∃ (nu : AppliedModelingLib.Matching.Assignment M W),
    Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w nu)
```

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.theorem2_side_optimal_stable_outcomes

Recorded source-to-Spec screening Verdict: matches

Reason: Theorem 2 says there is a stable outcome weakly preferred by every man to every other stable outcome and symmetrically for women. The repaired target now returns sourceMenOptimal and sourceWomenOptimal outcomes, each defined over complete source-stable marriages, exactly preserving the source's comparison class rather than the broader operational predicate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 356-357

Verbatim source input

THEOREM 3. No stable matching procedure for the general matching problem exists for which truthful revelation of preferences is a dominant strategy for all agents.

Expanded PaperInterface specification

```
¬∃ (mechanism :  
  (Roth82StableMatching.Theorem3Agent → Roth82StableMatching.Theorem3Agent → ℝ) →  
  (Roth82StableMatching.Theorem3Agent → Roth82StableMatching.Theorem3Agent → ℝ) →  
  AppliedModelingLib.Matching.Assignment Roth82StableMatching.Theorem3Agent Roth82StableMatching.Theorem3Agent),  
  Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure mechanism ∧  
  Roth82StableMatching.PaperInterface.truthfulForAllAgents mechanism
```

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure](#)
- [Roth82StableMatching.PaperInterface.truthfulForAllAgents](#)
- [Roth82StableMatching.Theorem3Agent](#)

Lean-elaborated claim roles

```
1. assumption: ∃ (mechanism :  
  (Roth82StableMatching.Theorem3Agent → Roth82StableMatching.Theorem3Agent → ℝ) →  
  (Roth82StableMatching.Theorem3Agent → Roth82StableMatching.Theorem3Agent → ℝ) →  
  AppliedModelingLib.Matching.Assignment Roth82StableMatching.Theorem3Agent Roth82StableMatching.Theorem3Agent),  
  Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure mechanism ∧  
  Roth82StableMatching.PaperInterface.truthfulForAllAgents mechanism  
2. conclusion: False
```

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.theorem3_no_stable_truthful_procedure

Recorded source-to-Spec screening Verdict: matches

Reason: Theorem 3 proves the general impossibility by exhibiting a three-men/three-women matching problem on which no stable procedure is truthful for all agents. The target formalizes precisely that fixed finite counterexample: no mechanism can simultaneously be source-stable on every legal profile and dominant-strategy truthful for both sides.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 438-439

Verbatim source input

THEOREM 4. Efficient matching procedures exist for which truthful revelation of preferences is a dominant strategy for every agent.

Expanded PaperInterface specification

$\forall (n : \mathbb{N}),$
 $\exists (\text{mechanism} : (\text{Fin } n \rightarrow \text{Fin } n \rightarrow \mathbb{R}) \rightarrow (\text{Fin } n \rightarrow \text{Fin } n \rightarrow \mathbb{R}) \rightarrow \text{AppliedModelingLib.Matching.Assignment } (\text{Fin } n) (\text{Fin } n)),$
Roth82StableMatching.PaperInterface.efficientMatchingProcedure mechanism $\wedge$
Roth82StableMatching.PaperInterface.truthfulForAllAgents mechanism

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.efficientMatchingProcedure](#)
- [Roth82StableMatching.PaperInterface.truthfulForAllAgents](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. conclusion: $\exists (\text{mechanism} : (\text{Fin } n \rightarrow \text{Fin } n \rightarrow \mathbb{R}) \rightarrow (\text{Fin } n \rightarrow \text{Fin } n \rightarrow \mathbb{R}) \rightarrow \text{AppliedModelingLib.Matching.Assignment } (\text{Fin } n) (\text{Fin } n)),$
Roth82StableMatching.PaperInterface.efficientMatchingProcedure mechanism $\wedge$
Roth82StableMatching.PaperInterface.truthfulForAllAgents mechanism

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.theorem4_efficient_truthful_procedure_exists

Recorded source-to-Spec screening Verdict: matches

Reason: Theorem 4 constructs efficient matching procedures with dominant truthful revelation for every agent. The target gives one such procedure on every finite equal-sided marriage market; efficientMatchingProcedure includes a complete Pareto-optimal outcome for every legal profile, and truthfulForAllAgents covers both sides' legal unilateral reports.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 468-475

Verbatim source input

THEOREM 5. In the matching procedure which always yields the optimal stable

outcome for a given one of the two sets of agents (i.e., for M or for W), truthful revelation is a dominant strategy for all the agents in that set.

[form-feed in source extraction]
624

ALVIN E. ROTH

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
[inst_3 : DecidableEq W] (val_m : M → W → ℝ) (val_w : W → M → ℝ),
Fintype.card M = Fintype.card W →
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
  (Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w
  (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w) ∧
  ∀ (m : M) (report_m : W → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
  (match
    (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m)
    val_w).m_match
    m with
    | none => 0
    | some w => val_m m w) ≤
  match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
  | none => 0
  | some w => val_m m w) ∧
  Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w
  (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w) ∧
  ∀ (w : W) (report_w : M → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w) →
  (match
    (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m
    (Function.update val_w w report_w)).w_match
    w with
    | none => 0
    | some m => val_w w m) ≤
  match (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w).w_match w with
  | none => 0
  | some m => val_w w m
```

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.menDeferredAcceptance](#)
- [Roth82StableMatching.PaperInterface.sourceMenOptimal](#)
- [Roth82StableMatching.PaperInterface.sourceWomenOptimal](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)
- [Roth82StableMatching.PaperInterface.womenDeferredAcceptance](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: Fintype M
4. parameter: Fintype W
5. parameter: DecidableEq M
6. parameter: DecidableEq W
7. parameter: M → W → ℝ
8. parameter: W → M → ℝ
9. assumption: Fintype.card M = Fintype.card W
10. assumption: Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w
11. conclusion: (Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w
  (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w) ∧
  ∀ (m : M) (report_m : W → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
  (match
    (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m) val_w).m_match
    m with
    | none => 0
    | some w => val_m m w) ≤
  match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
  | none => 0
```

```

| some w => val_m m w) ∧
Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w
(Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w) ∧
∀ (w : W) (report_w : M → ℝ),
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w) →
(match
  (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m
    (Function.update val_w w report_w)).w_match
    w with
  | none => 0
  | some m => val_w w m) ≤
match (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w).w_match w with
| none => 0
| some m => val_w w m

```

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.theorem5_optimal_side_truthful

Recorded source-to-Spec screening Verdict: matches

Reason: Theorem 5 states that the procedure selecting the optimal stable outcome for either side makes truth-telling dominant for all agents on that side. The target pairs the men- and women-proposing deferred-acceptance outcomes with their respective source-optimality and one-side dominant-strategy inequalities on legal reports.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 477-535

Verbatim source input

COROLLARY 5.1. In the matching procedure which always yields the optimal stable outcome for a given set of agents, the agents in the other set have no incentive to misrepresent their first choice.

Note that both results are phrased in terms of "the" matching procedure which yields a particular outcome. There are obviously different procedures which yield the same outcome, but from the point of view of incentives such procedures are equivalent, and can be regarded as a single procedure. Note also that
↪ Theorem 5 implies that, although an agent can in general change the set of stable outcomes by misrepresenting his preferences, no agent can manipulate his preferences in such a way that he prefers his best outcome in the altered set of stable outcomes to his best outcome in the

original set of stable outcomes. The next section is devoted to the proof of these results, which is somewhat more complex than the proof of the earlier results. presents some additional results about the structure of the set of stable outcome

which arise in the course of the proofs.

5. Proofs of Theorem 5 and its corollary. In this section it will be shown that the repeated proposal procedure G used in the proof of Theorem 1 has the properties stated in Theorem 5 and Corollary 5.1. As discussed in section 2, to establish the results for the general matching problem with strict preferences, it will be sufficient

demonstrate them for the marriage problem. Throughout this section, therefore, the sets of agents will be $M = \{m_1, \dots, m_n\}$ and $W = \{w_1, \dots, w_n\}$, the true preferences of the players will be given by the arbitrary preference profile P , and $x = g(P)$ will denote the outcome which results from the repeated proposal procedure when the true preferences are stated.

To prove Theorem 5, we need to show that truthful revelation is a dominant strategy

for each m_i in M . Since the (true) preference profile P is arbitrary, it will be sufficient to show that, if P' is a preference profile which differs from P only in that $P'(m_i)$, say

replaces m_i 's true preferences $P(m_i)$, then m_i doesn't prefer the outcome $y = g(P')$ to $x = g(P)$. That is, we need to show that no successful misrepresentation

of preferences is possible, where a misrepresentation $P'(m_i)$ is defined to be

successful misrepresentation by m_i if $y(m_i)P(m_i)x(m_i)$. That is, a misrepresentation $P'(m_i)$ is successful if m_i prefers (according to his true preferences) the partner

$y(m_i)$ he's matched with when he misrepresents his preferences to the partner $x(m_i)$ he's matched with when he states his true preferences. (Throughout this section $y = g(P')$ will denote an outcome resulting from misrepresentation by m_i .)

We first show that only a certain kind of simple misrepresentation need

be considered, since if any successful manipulation is possible, then it can be achieved by a simple misrepresentation. Specifically, if $P'(m_i)$ is an arbitrary misrepresentation discussed above, then an equivalent simple misrepresentation $P''(m_i)$ is one in which

$y(m_i)P''(m_i)w_j$ for all $w_j \neq y(m_i)$. That is, $P''(m_i)$ is a preference relation which has $y(m_i)$ as the most preferred match of m_i . The justification for calling $P''(m_i)$ an equivalent misrepresentation to $P'(m_i)$ is given by the following lemma, which states

that m_i will end up matched to the same partner, $y(m_i)$, whether he misrepresents using

$P'(m_i)$ or $P''(m_i)$. (The preference profile P'' , of course, denotes the one which differs from P only in that $P''(m_i)$ replaces $P(m_i)$.)

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
[inst_3 : DecidableEq W] [Nonempty M] [Nonempty W],
Fintype.card M = Fintype.card W →
(∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w
(Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w) ∧
∀ (w : W) (report_w : M → ℝ),
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w) →
∃ (report_w' : M → ℝ),
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w') ∧
(∀ (mstar : M),
(∀ (m : M), m ≠ mstar → val_w w m < val_w w mstar) →
∀ (m : M), m ≠ mstar → report_w' m < report_w' mstar) ∧
match
(Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m
(Function.update val_w w report_w)).w_match
with
| none => 0
| some m => val_w w m ≤
match
(Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m
(Function.update val_w w report_w')).w_match
```

```

      w with
      | none => 0
      | some m => val_w w m) ∧
  ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
  Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w
  (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w) ∧
  ∀ (m : M) (report_m : W → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
  ∃ (report_m' : W → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m') val_w ∧
  (∀ (wstar : W),
    (∀ (w : W), w ≠ wstar → val_m m w < val_m m wstar) →
    ∀ (w : W), w ≠ wstar → report_m' w < report_m' wstar) ∧
  (match
    (Roth82StableMatching.PaperInterface.womenDeferredAcceptance
      (Function.update val_m m report_m) val_w).m_match
    with
    | none => 0
    | some w => val_m m w) ≤
  (match
    (Roth82StableMatching.PaperInterface.womenDeferredAcceptance
      (Function.update val_m m report_m') val_w).m_match
    with
    | none => 0
    | some w => val_m m w)

```

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.menDeferredAcceptance](#)
- [Roth82StableMatching.PaperInterface.sourceMenOptimal](#)
- [Roth82StableMatching.PaperInterface.sourceWomenOptimal](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)
- [Roth82StableMatching.PaperInterface.womenDeferredAcceptance](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: Fintype M
4. parameter: Fintype W
5. parameter: DecidableEq M
6. parameter: DecidableEq W
7. assumption: Nonempty M
8. assumption: Nonempty W
9. assumption: Fintype.card M = Fintype.card W
10. conclusion: (∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
  Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w
  (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w) ∧
  ∀ (w : W) (report_w : M → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w) →
  ∃ (report_w' : M → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w') ∧
  (∀ (mstar : M),
    (∀ (m : M), m ≠ mstar → val_w w m < val_w w mstar) →
    ∀ (m : M), m ≠ mstar → report_w' m < report_w' mstar) ∧
  (match
    (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m
      (Function.update val_w w report_w)).w_match
    with
    | none => 0
    | some m => val_w w m) ≤
  (match
    (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m
      (Function.update val_w w report_w')).w_match
    with
    | none => 0
    | some m => val_w w m) ∧
  ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
  Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w
  (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w) ∧
  ∀ (m : M) (report_m : W → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
  ∃ (report_m' : W → ℝ),
  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m') val_w ∧
  (∀ (wstar : W),
    (∀ (w : W), w ≠ wstar → val_m m w < val_m m wstar) →
    ∀ (w : W), w ≠ wstar → report_m' w < report_m' wstar) ∧
  (match
    (Roth82StableMatching.PaperInterface.womenDeferredAcceptance (Function.update val_m m report_m)
      val_w).m_match
    with
    | none => 0
    | some w => val_m m w) ≤
  (match
    (Roth82StableMatching.PaperInterface.womenDeferredAcceptance (Function.update val_m m report_m')
      val_w).m_match
    with
    | none => 0
    | some w => val_m m w)

```

```

match
  (Roth82StableMatching.PaperInterface.womenDeferredAcceptance (Function.update val_m m report_m')
    val_w).m_match
  m with
| none => 0
| some w => val_m m w

```

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.corollary5_1_first_choice_need_not_be_misrepresented

Recorded source-to-Spec screening Verdict: matches

Reason: Corollary 5.1 says the non-optimal side need not misreport its first choice. The target records that, for either orientation, every legal unilateral report has a legal weakly better replacement which preserves any true first choice as first; this is the source's no-gain-from-changing-first-choice statement for the procedure selecting the opposite side's optimal stable outcome.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 537

Verbatim source input

LEMMA 1. If $y = g(P')$ and $z = g(P'')$ then $z(mi) = y(mi)$.

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
[inst_3 : DecidableEq W] (val_m : M → W → ℝ) (val_w : W → M → ℝ) (m : M) (report_m simple_report_m : W → ℝ)
(wstar : W),
Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m simple_report_m) val_w →
(Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m) val_w).m_match m =
some wstar →
(∀ (w : W), w ≠ wstar → simple_report_m w < simple_report_m wstar) →
(Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m)
val_w).m_match
m =
(Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m) val_w).m_match
m
```

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.menDeferredAcceptance](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: Fintype M
4. parameter: Fintype W
5. parameter: DecidableEq M
6. parameter: DecidableEq W
7. parameter: M → W → ℝ
8. parameter: W → M → ℝ
9. parameter: M
10. parameter: W → ℝ
11. parameter: W → ℝ
12. parameter: W
13. assumption: Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w
14. assumption: Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m simple_report_m) val_w
15. assumption: (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m) val_w).m_match m =
some wstar
16. assumption: ∀ (w : W), w ≠ wstar → simple_report_m w < simple_report_m wstar
17. conclusion: (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m) val_w).m_match m =
(Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m) val_w).m_match m
```

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.lemma1_simple_report_same_partner

Recorded source-to-Spec screening Verdict: matches

Reason: Lemma 1 (pp. 621–22) compares the reported profile P' with the simple profile P'' , both of which are preference profiles, and concludes $z(mi)=y(mi)$. The target now expressly requires strict legal domains for both updated reports, assumes that P' produces $wstar$ and that P'' ranks $wstar$ first, and concludes the same match for mi .

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 564-567

Verbatim source input

LEMMA 2. If $P'(mi)$ is a simple misrepresentation such that $y = g(P')$ and either $y(m)P(mi)x(mi)$ or $y(mi) = x(mi)$ then, for each m_j in M , either $y(m_j)P(m_j)x(m_j)$ or $y(m_j) = x(m_j)$.

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
[inst_3 : DecidableEq W] (val_m : M → W → ℝ) (val_w : W → M → ℝ) (m : M) (simple_report_m : W → ℝ) (ystar : W),
Fintype.card M = Fintype.card W →
  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m simple_report_m) val_w →
  (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m)
    val_w).m_match
    m =
  some ystar →
  ((match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
  | none => 0
  | some w => val_m m w) ≤
  match
    (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m)
      val_w).m_match
      m with
  | none => 0
  | some w => val_m m w) →
  Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst simple_report_m ystar →
  ∀ (m' : M),
  (match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m' with
  | none => 0
  | some w => val_m m' w) ≤
  match
    (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m)
      val_w).m_match
      m' with
  | none => 0
  | some w => val_m m' w
```

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst](#)
- [Roth82StableMatching.PaperInterface.menDeferredAcceptance](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)

Lean-elaborated claim roles

1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: Fintype M
4. parameter: Fintype W
5. parameter: DecidableEq M
6. parameter: DecidableEq W
7. parameter: M → W → ℝ
8. parameter: W → M → ℝ
9. parameter: M
10. parameter: W → ℝ
11. parameter: W
12. assumption: Fintype.card M = Fintype.card W
13. assumption: Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w
14. assumption: Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m simple_report_m) val_w
15. assumption: (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m) val_w).m_match m = some ystar
16. assumption: (match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
| none => 0
| some w => val_m m w) ≤
match
 (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m) val_w).m_match
 m with
| none => 0
| some w => val_m m w
17. assumption: Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst simple_report_m ystar
18. parameter: M
19. conclusion: (match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m' with
| none => 0
| some w => val_m m' w) ≤
match
 (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m) val_w).m_match
 m' with
| none => 0
| some w => val_m m' w

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.lemma2_simple_report_harms_no_man

Recorded source-to-Spec screening Verdict: matches

Reason: Lemma 2 states that a simple unilateral report which leaves m_i weakly better off cannot make any man worse off. The target uses the men-proposing procedure, legal true and changed profiles, the simple-report/top-partner condition, and weak comparison of every man's true utility, matching that conditional source conclusion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 678-684

Verbatim source input

THEOREM 6. No feasible outcome is strictly preferred by all m_i in M to the outcome $g(P)$.

The fact that no stable outcome is strictly preferred to $g(P)$ isn't news: Theorem 2 gives a stronger result. What Theorem 6 says is that in fact, $g(P)$ is weakly Pareto optimal from the point of view of the men, with respect to any possible outcome. The

example above shows that this can't be strengthened to strong Pareto optimality.

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
[inst_3 : DecidableEq W] [Nonempty M] (val_m : M → W → ℝ) (val_w : W → M → ℝ),
Fintype.card M = Fintype.card W →
Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
¬∃ (nu : AppliedModelingLib.Matching.Assignment M W),
Roth82StableMatching.PaperInterface.completeMarriageOutcome nu ∧
∀ (m : M),
(match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
| none => 0
| some w => val_m m w) <
match nu.m_match m with
| none => 0
| some w => val_m m w
```

Retained governing declarations

- [AppliedModelingLib.Matching.Assignment](#)
- [Roth82StableMatching.PaperInterface.completeMarriageOutcome](#)
- [Roth82StableMatching.PaperInterface.menDeferredAcceptance](#)
- [Roth82StableMatching.PaperInterface.strictMarriageDomain](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: Fintype M
4. parameter: Fintype W
5. parameter: DecidableEq M
6. parameter: DecidableEq W
7. assumption: Nonempty M
8. parameter: M → W → ℝ
9. parameter: W → M → ℝ
10. assumption: Fintype.card M = Fintype.card W
11. assumption: Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w
12. conclusion: ¬∃ (nu : AppliedModelingLib.Matching.Assignment M W),
Roth82StableMatching.PaperInterface.completeMarriageOutcome nu ∧
∀ (m : M),
(match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
| none => 0
| some w => val_m m w) <
match nu.m_match m with
| none => 0
| some w => val_m m w
```

Lean proof endpoint (built separately)

Roth82StableMatching.PaperInterface.theorem6_no_outcome_strictly_better_for_all_men

Recorded source-to-Spec screening Verdict: matches

Reason: Theorem 6 says no feasible outcome is strictly preferred by all men to $g(P)$, the men-proposing outcome. The target excludes exactly a complete feasible assignment that gives every man a strict true-utility improvement over menDeferredAcceptance, under the source's finite strict marriage assumptions.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 713-714

Verbatim source input

THEOREM 7. No stable matching procedure exists which never gives any agent an incentive to misrepresent his k th choice, for $k \neq 1$.

Expanded PaperInterface specification

$\forall (k : \mathbb{N}),$
 $1 < k \rightarrow$
 $\exists (n : \mathbb{N}), \text{Roth82StableMatching.paper_no_stable_procedure_avoids_strict_kth_choice_manipulation_on } (\text{Fin } n) (\text{Fin } n) k$

Retained governing declarations

- [Roth82StableMatching.paper_no_stable_procedure_avoids_strict_kth_choice_manipulation_on](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. assumption: $1 < k$
3. conclusion: $\exists (n : \mathbb{N}), \text{Roth82StableMatching.paper_no_stable_procedure_avoids_strict_kth_choice_manipulation_on } (\text{Fin } n) (\text{Fin } n) k$

Lean proof endpoint (built separately)

`Roth82StableMatching.PaperInterface.theorem7_later_choice_manipulation_unavoidable`

Recorded source-to-Spec screening Verdict: matches

Reason: Theorem 7 says that for every k th choice with k not first, no stable procedure can avoid incentives to misrepresent it; the proof states $k > 1$. The target uses the equivalent natural-number condition $1 < k$ and asserts a finite witness instance for each such rank, matching the source's impossibility form.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation